

Mini Project: Unsupervised Learning of Cancer Data

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```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.1      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.3      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr      1.2.2
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(ggplot2)
```

Data import

```
fna.data <- "https://bioboot.github.io/bimm143_S20/class-material/WisconsinCancer.csv"
wisc.df <- read.csv(fna.data, row.names = 1)
head(wisc.df, 4)
```

	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean
842302	M	17.99	10.38	122.80	1001.0
842517	M	20.57	17.77	132.90	1326.0
84300903	M	19.69	21.25	130.00	1203.0

84348301	M	11.42	20.38	77.58	386.1
	smoothness_mean	compactness_mean	concavity_mean	concave.points_mean	
842302	0.11840	0.27760	0.3001	0.14710	
842517	0.08474	0.07864	0.0869	0.07017	
84300903	0.10960	0.15990	0.1974	0.12790	
84348301	0.14250	0.28390	0.2414	0.10520	
	symmetry_mean	fractal_dimension_mean	radius_se	texture_se	perimeter_se
842302	0.2419	0.07871	1.0950	0.9053	8.589
842517	0.1812	0.05667	0.5435	0.7339	3.398
84300903	0.2069	0.05999	0.7456	0.7869	4.585
84348301	0.2597	0.09744	0.4956	1.1560	3.445
	area_se	smoothness_se	compactness_se	concavity_se	concave.points_se
842302	153.40	0.006399	0.04904	0.05373	0.01587
842517	74.08	0.005225	0.01308	0.01860	0.01340
84300903	94.03	0.006150	0.04006	0.03832	0.02058
84348301	27.23	0.009110	0.07458	0.05661	0.01867
	symmetry_se	fractal_dimension_se	radius_worst	texture_worst	
842302	0.03003	0.006193	25.38	17.33	
842517	0.01389	0.003532	24.99	23.41	
84300903	0.02250	0.004571	23.57	25.53	
84348301	0.05963	0.009208	14.91	26.50	
	perimeter_worst	area_worst	smoothness_worst	compactness_worst	
842302	184.60	2019.0	0.1622	0.6656	
842517	158.80	1956.0	0.1238	0.1866	
84300903	152.50	1709.0	0.1444	0.4245	
84348301	98.87	567.7	0.2098	0.8663	
	concavity_worst	concave.points_worst	symmetry_worst		
842302	0.7119	0.2654	0.4601		
842517	0.2416	0.1860	0.2750		
84300903	0.4504	0.2430	0.3613		
84348301	0.6869	0.2575	0.6638		
	fractal_dimension_worst				
842302	0.11890				
842517	0.08902				
84300903	0.08758				
84348301	0.17300				

- Removing diagnosis from data for PCA and clustering

```
wisc.data <- wisc.df[, -1]

diagnosis <- as.factor(wisc.df$diagnosis)
```

```
dim(wisc.data)
```

```
[1] 569 30
```

```
table(diagnosis)
```

```
diagnosis  
  B    M  
357 212
```

Exploratory data analysis

Q1. How many observations in the dataset?

```
nrow(wisc.data)
```

```
[1] 569
```

There are 569 observations.

Q2. How many observations contain a malignant diagnosis?

```
table(diagnosis)
```

```
diagnosis  
  B    M  
357 212
```

There 212 malignant diagnosis.

Q3. How many variables/features suffixed with __mean?

```
length(grep("__mean$", names(wisc.data)))
```

```
[1] 10
```

There are 10 variables ending in __mean.

Principal Component Analysis

- Checking means and standard deviations of variables

```
colMeans(wisc.data)
```

radius_mean	texture_mean	perimeter_mean
1.412729e+01	1.928965e+01	9.196903e+01
area_mean	smoothness_mean	compactness_mean
6.548891e+02	9.636028e-02	1.043410e-01
concavity_mean	concave.points_mean	symmetry_mean
8.879932e-02	4.891915e-02	1.811619e-01
fractal_dimension_mean	radius_se	texture_se
6.279761e-02	4.051721e-01	1.216853e+00
perimeter_se	area_se	smoothness_se
2.866059e+00	4.033708e+01	7.040979e-03
compactness_se	concavity_se	concave.points_se
2.547814e-02	3.189372e-02	1.179614e-02
symmetry_se	fractal_dimension_se	radius_worst
2.054230e-02	3.794904e-03	1.626919e+01
texture_worst	perimeter_worst	area_worst
2.567722e+01	1.072612e+02	8.805831e+02
smoothness_worst	compactness_worst	concavity_worst
1.323686e-01	2.542650e-01	2.721885e-01
concave.points_worst	symmetry_worst	fractal_dimension_worst
1.146062e-01	2.900756e-01	8.394582e-02

```
apply(wisc.data, 2, sd)
```

radius_mean	texture_mean	perimeter_mean
3.524049e+00	4.301036e+00	2.429898e+01
area_mean	smoothness_mean	compactness_mean
3.519141e+02	1.406413e-02	5.281276e-02
concavity_mean	concave.points_mean	symmetry_mean
7.971981e-02	3.880284e-02	2.741428e-02
fractal_dimension_mean	radius_se	texture_se
7.060363e-03	2.773127e-01	5.516484e-01
perimeter_se	area_se	smoothness_se
2.021855e+00	4.549101e+01	3.002518e-03
compactness_se	concavity_se	concave.points_se
1.790818e-02	3.018606e-02	6.170285e-03
symmetry_se	fractal_dimension_se	radius_worst

8.266372e-03	2.646071e-03	4.833242e+00
texture_worst	perimeter_worst	area_worst
6.146258e+00	3.360254e+01	5.693570e+02
smoothness_worst	compactness_worst	concavity_worst
2.283243e-02	1.573365e-01	2.086243e-01
concave.points_worst	symmetry_worst	fractal_dimension_worst
6.573234e-02	6.186747e-02	1.806127e-02

There are many scales, therefore scaling before PCA is a must.

```
wisc.pr <- prcomp(wisc.data, scale. = TRUE)
summary(wisc.pr)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	3.6444	2.3857	1.67867	1.40735	1.28403	1.09880	0.82172
Proportion of Variance	0.4427	0.1897	0.09393	0.06602	0.05496	0.04025	0.02251
Cumulative Proportion	0.4427	0.6324	0.72636	0.79239	0.84734	0.88759	0.91010
	PC8	PC9	PC10	PC11	PC12	PC13	PC14
Standard deviation	0.69037	0.6457	0.59219	0.5421	0.51104	0.49128	0.39624
Proportion of Variance	0.01589	0.0139	0.01169	0.0098	0.00871	0.00805	0.00523
Cumulative Proportion	0.92598	0.9399	0.95157	0.9614	0.97007	0.97812	0.98335
	PC15	PC16	PC17	PC18	PC19	PC20	PC21
Standard deviation	0.30681	0.28260	0.24372	0.22939	0.22244	0.17652	0.1731
Proportion of Variance	0.00314	0.00266	0.00198	0.00175	0.00165	0.00104	0.0010
Cumulative Proportion	0.98649	0.98915	0.99113	0.99288	0.99453	0.99557	0.9966
	PC22	PC23	PC24	PC25	PC26	PC27	PC28
Standard deviation	0.16565	0.15602	0.1344	0.12442	0.09043	0.08307	0.03987
Proportion of Variance	0.00091	0.00081	0.0006	0.00052	0.00027	0.00023	0.00005
Cumulative Proportion	0.99749	0.99830	0.9989	0.99942	0.99969	0.99992	0.99997
	PC29	PC30					
Standard deviation	0.02736	0.01153					
Proportion of Variance	0.00002	0.00000					
Cumulative Proportion	1.00000	1.00000					

Q4. What proportion of the original variance is captured by PC1?

PC1 captures 44.27% of the original variance.

Q5. How many PCs required to describe 70% of the original variance?

```
pve <- wisc.pr$sdev^2 / sum(wisc.pr$sdev^2)
which(cumsum(pve) >= 0.70)[1]
```

```
[1] 3
```

A minimum of 3 principal components required to describe 70% of original variance.

Q6. How many PCs are required to describe 90% of the original variance?

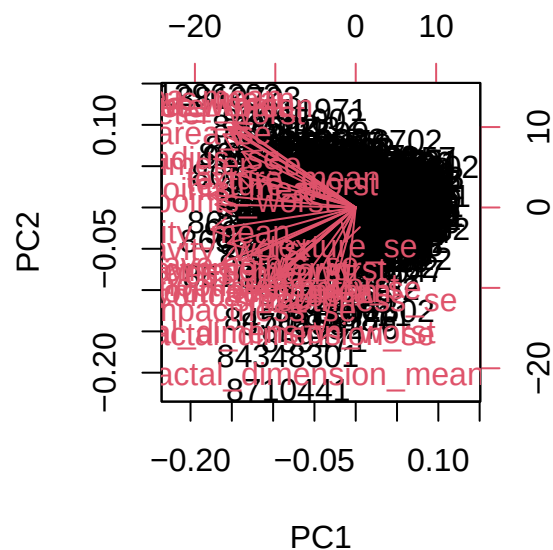
```
which(cumsum(pve) >= 0.90)[1]
```

```
[1] 7
```

A minimum of 7 principal components required to describe 90% of the original variance.

Interpreting PCA results

```
biplot(wisc.pr)
```



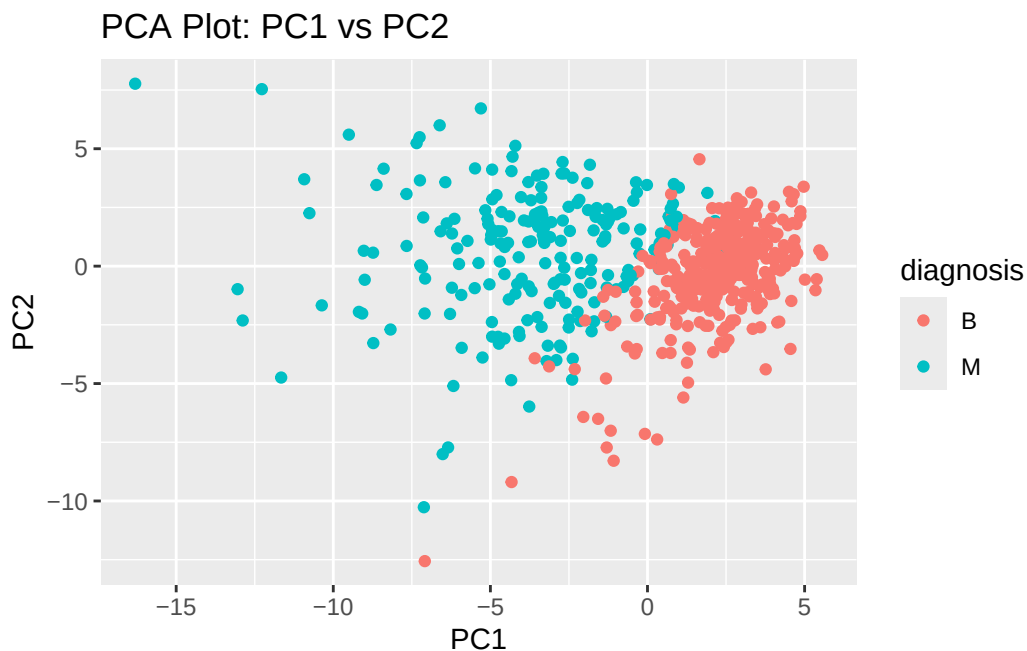
Q7. What stands out about this plot?

The biplot is difficult to interpret because there are many observations and variables plotted at once. The row names overlap heavily, making it hard to clearly see patterns or separation between samples.

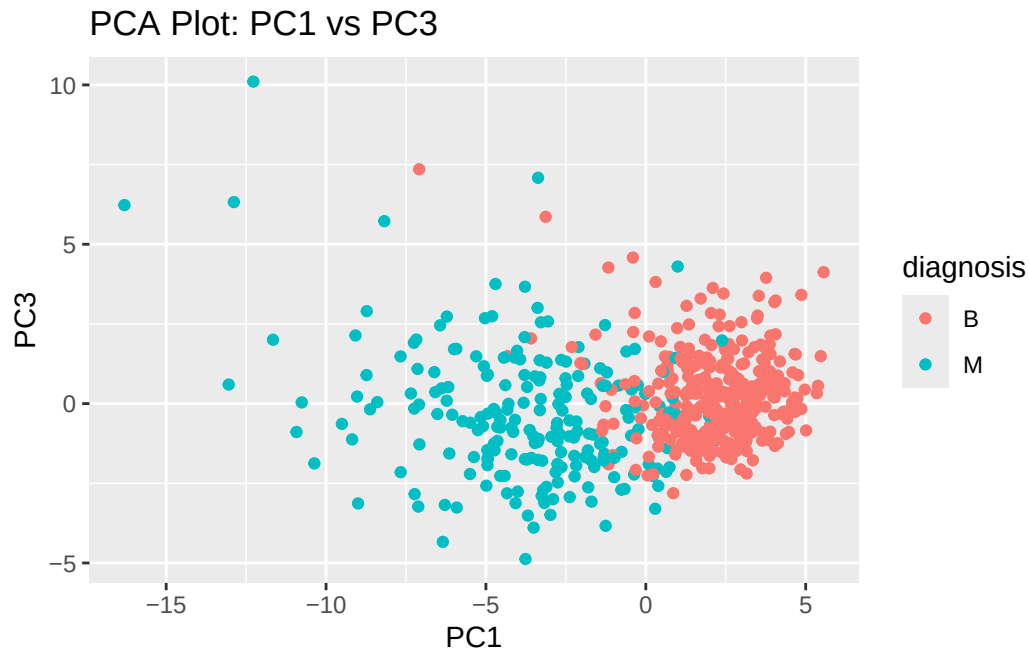
Cleaning up the plot:

```
pc_df <- as.data.frame(wisc.pr$x)
pc_df$diagnosis <- diagnosis

ggplot(pc_df, aes(PC1, PC2, color = diagnosis)) +
  geom_point() +
  labs(title = "PCA Plot: PC1 vs PC2",
       x = "PC1",
       y = "PC2")
```



```
ggplot(pc_df, aes(PC1, PC3, color = diagnosis)) +
  geom_point() +
  labs(title = "PCA Plot: PC1 vs PC3",
       x = "PC1",
       y = "PC3")
```



Q8. What do you notice about these plots?

Both PCA plots show that PC1 separates many malignant and benign samples. The PC1 vs PC2 plot shows clearer separation than the biplot, while the PC1 vs PC3 plot still shows separation but with more overlap.

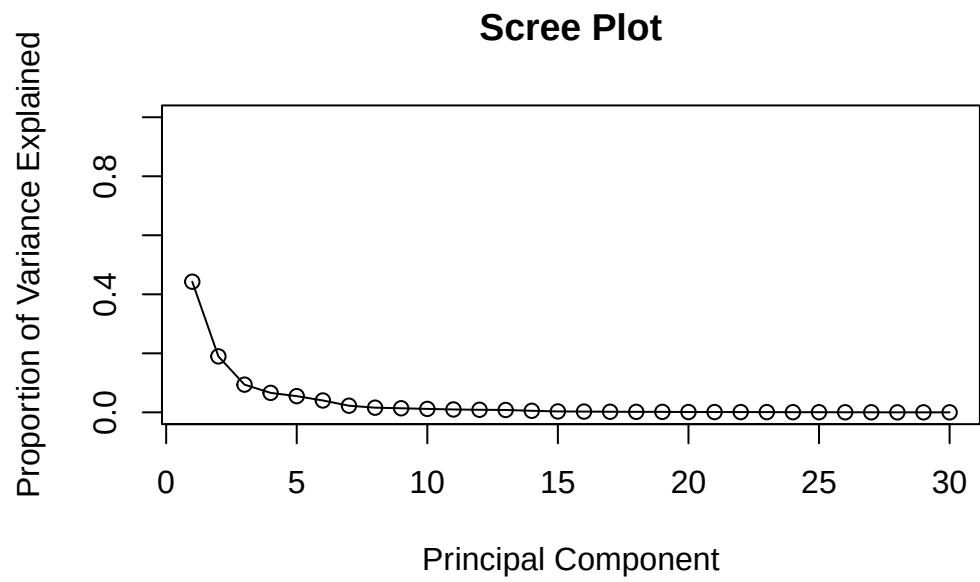
Variance Explained

```
pr.var <- wisc.pr$sdev^2
head(pr.var)
```

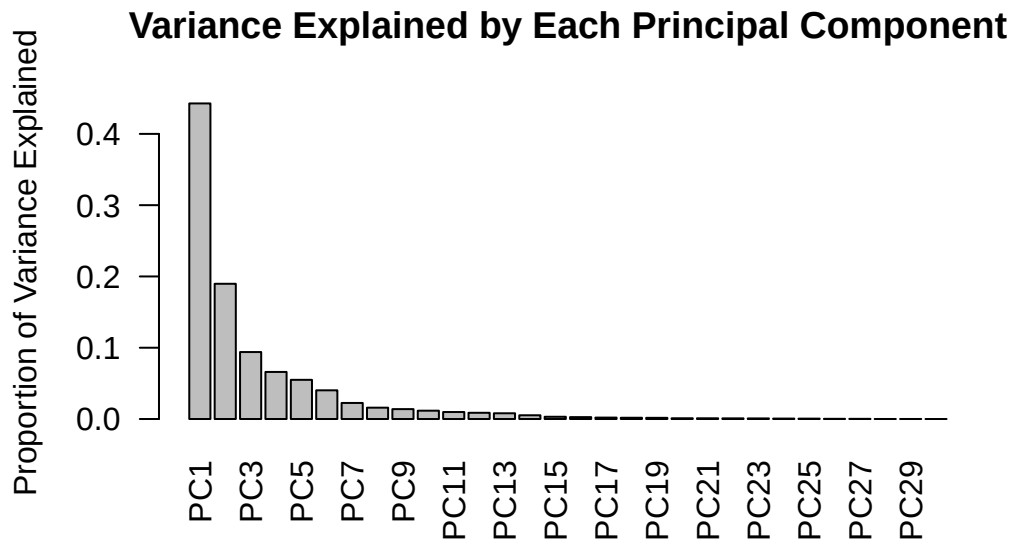
```
[1] 13.281608  5.691355  2.817949  1.980640  1.648731  1.207357
```

```
pve <- pr.var / sum(pr.var)

plot(pve,
     xlab = "Principal Component",
     ylab = "Proportion of Variance Explained",
     ylim = c(0, 1),
     type = "o",
     main = "Scree Plot")
```

```
barplot(pve,  
  ylab = "Proportion of Variance Explained",  
  names.arg = paste0("PC", 1:length(pve)),  
  las = 2,  
  main = "Variance Explained by Each Principal Component")
```



Most of the variance is explained by the first few principal components. PC1 explains the most variance, and the amount explained decreases after each additional component.

PCA loadings

```
wisc.pr$rotation["concave.points_mean", 1]
```

```
[1] -0.2608538
```

```
sort(abs(wisc.pr$rotation[, 1]), decreasing = TRUE)
```

concave.points_mean	concavity_mean	concave.points_worst
0.26085376	0.25840048	0.25088597
compactness_mean	perimeter_worst	concavity_worst
0.23928535	0.23663968	0.22876753
radius_worst	perimeter_mean	area_worst
0.22799663	0.22753729	0.22487053
area_mean	radius_mean	perimeter_se
0.22099499	0.21890244	0.21132592
compactness_worst	radius_se	area_se
0.21009588	0.20597878	0.20286964

concave.points_se	compactness_se	concavity_se
0.18341740	0.17039345	0.15358979
smoothness_mean	symmetry_mean	fractal_dimension_worst
0.14258969	0.13816696	0.13178394
smoothness_worst	symmetry_worst	texture_worst
0.12795256	0.12290456	0.10446933
texture_mean	fractal_dimension_se	fractal_dimension_mean
0.10372458	0.10256832	0.06436335
symmetry_se	texture_se	smoothness_se
0.04249842	0.01742803	0.01453145

Q9. What is the loading for `concave.points_mean` on PC1? Are there larger contributions?

The loading for `concave.points_mean` on PC1 is approximately -0.26085. Based on the absolute values of the loadings, this is one of the strongest contributors to PC1 and appears to be the largest absolute contribution.

Hierarchical clustering

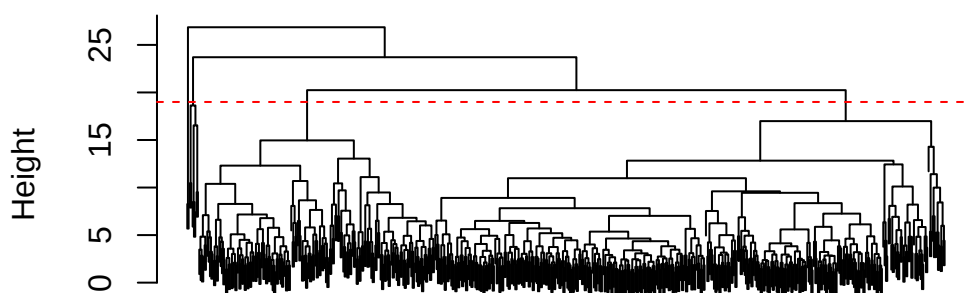
```
data.scaled <- scale(wisc.data)

data.dist <- dist(data.scaled)

wisc.hclust <- hclust(data.dist, method = "complete")

plot(wisc.hclust, labels = FALSE, main = "Hierarchical Clustering")
abline(h = 19, col = "red", lty = 2)
```

Hierarchical Clustering



```
data.dist  
hclust (*, "complete")
```

Q10. What height gives 4 clusters?

A height of 19 gives 4 clusters.

```
wisc.hclust.clusters <- cutree(wisc.hclust, k = 4)  
  
table(wisc.hclust.clusters, diagnosis)
```

	diagnosis	
wisc.hclust.clusters	B	M
1	12	165
2	2	5
3	343	40
4	0	2

Using 4 clusters, cluster 1 is mostly malignant and cluster 3 is mostly benign. However, there are still some misclassified samples.

Trying different clustering methods

```

hc.single <- hclust(data.dist, method = "single")
hc.average <- hclust(data.dist, method = "average")
hc.complete <- hclust(data.dist, method = "complete")
hc.ward <- hclust(data.dist, method = "ward.D2")

table(cutree(hc.single, k = 2), diagnosis)

```

```

diagnosis
  B   M
1 357 210
2   0   2

```

```

table(cutree(hc.average, k = 2), diagnosis)

```

```

diagnosis
  B   M
1 357 209
2   0   3

```

```

table(cutree(hc.complete, k = 2), diagnosis)

```

```

diagnosis
  B   M
1 357 210
2   0   2

```

```

table(cutree(hc.ward, k = 2), diagnosis)

```

```

diagnosis
  B   M
1  20 164
2 337  48

```

Q12. Which method provides favorite results?

ward.D2 because it is balances and considerate of biological samples. It minimizes variation within clusters which helps with this dataset that is trying to group similar tumor samples together.

Combining PCA and clustering

Since 7 PCs explain about 90% of variance, here I use the first 7 PCs to cluster.

```
wisc.pr.hclust <- hclust(dist(wisc.pr$x[, 1:7]), method = "ward.D2")  
  
plot(wisc.pr.hclust,  
      labels = FALSE,  
      main = "Hierarchical Clustering on First 7 PCs")
```

Hierarchical Clustering on First 7 PCs



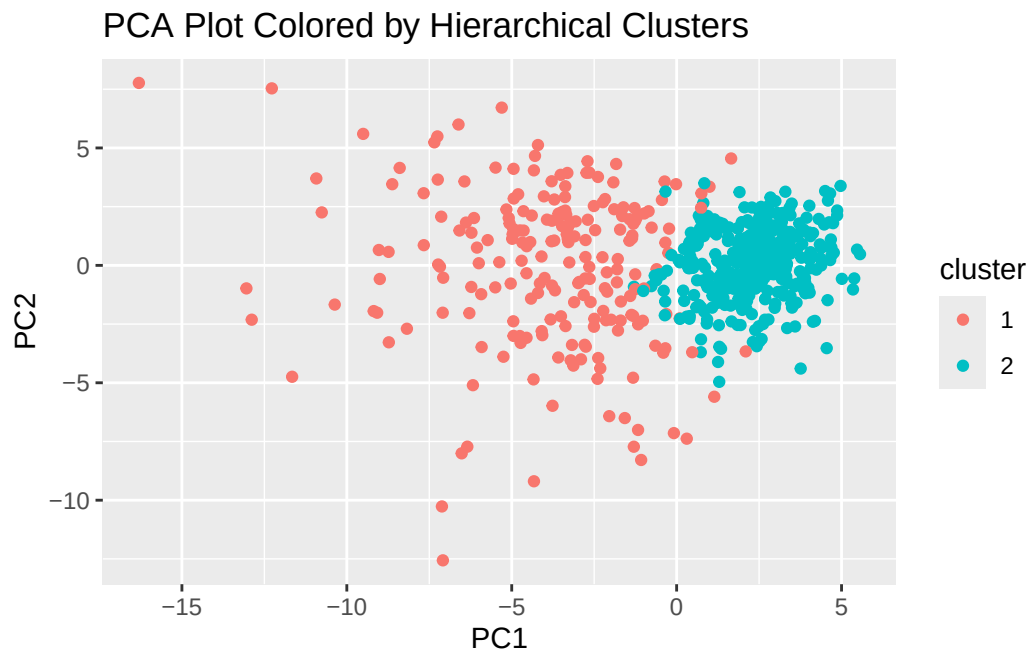
```
dist(wisc.pr$x[, 1:7])  
hclust (*, "ward.D2")
```

```
wisc.pr.hclust.clusters <- cutree(wisc.pr.hclust, k = 2)  
  
table(wisc.pr.hclust.clusters, diagnosis)
```

```
              diagnosis  
wisc.pr.hclust.clusters  B    M  
1      28 188  
2     329  24
```

```
pc_df$cluster <- as.factor(wisc.pr.hclust.clusters)  
  
ggplot(pc_df, aes(PC1, PC2, color = cluster)) +
```

```
geom_point() +
labs(title = "PCA Plot Colored by Hierarchical Clusters",
     x = "PC1",
     y = "PC2")
```



Q13. How well does the PCA-based clustering separate malignant and benign samples?

The PCA-based clustering separates the diagnoses fairly well. One cluster is mostly malignant, while the other is mostly benign. However, the separation is not perfect because some malignant samples are grouped with benign samples and some benign samples are grouped with malignant samples.

Q14. How well do the original hierarchical clustering models separate diagnoses compared with PCA-based clustering?

```
table(wisc.hclust.clusters, diagnosis)
```

	diagnosis	
wisc.hclust.clusters	B	M
1	12	165
2	2	5
3	343	40
4	0	2

```
table(wisc.pr.hclust.clusters, diagnosis)
```

```

              diagnosis
wisc.pr.hclust.clusters  B  M
1    28 188
2   329  24

```

The PCA-based clustering gives a cleaner two-cluster result compared with clustering on all scaled original variables. Using PCA first reduces noise and focuses the clustering on the strongest patterns in the data.

Prediction with new samples

```

url <- "https://tinyurl.com/new-samples-CSV"
new <- read.csv(url)

npc <- predict(wisc.pr, newdata = new)

npc

```

```

      PC1      PC2      PC3      PC4      PC5      PC6      PC7
[1,]  2.576616 -3.135913  1.3990492 -0.7631950  2.781648 -0.8150185 -0.3959098
[2,] -4.754928 -3.009033 -0.1660946 -0.6052952 -1.140698 -1.2189945  0.8193031
      PC8      PC9      PC10      PC11      PC12      PC13      PC14
[1,] -0.2307350  0.1029569 -0.9272861  0.3411457  0.375921  0.1610764  1.187882
[2,] -0.3307423  0.5281896 -0.4855301  0.7173233 -1.185917  0.5893856  0.303029
      PC15      PC16      PC17      PC18      PC19      PC20
[1,]  0.3216974 -0.1743616 -0.07875393 -0.11207028 -0.08802955 -0.2495216
[2,]  0.1299153  0.1448061 -0.40509706  0.06565549  0.25591230 -0.4289500
      PC21      PC22      PC23      PC24      PC25      PC26
[1,]  0.1228233  0.09358453  0.08347651  0.1223396  0.02124121  0.078884581
[2,] -0.1224776  0.01732146  0.06316631 -0.2338618 -0.20755948 -0.009833238
      PC27      PC28      PC29      PC30
[1,]  0.220199544 -0.02946023 -0.015620933  0.005269029
[2,] -0.001134152  0.09638361  0.002795349 -0.019015820

```

```

plot(wisc.pr$x[, 1:2],
     col = wisc.pr.hclust.clusters,
     xlab = "PC1",

```



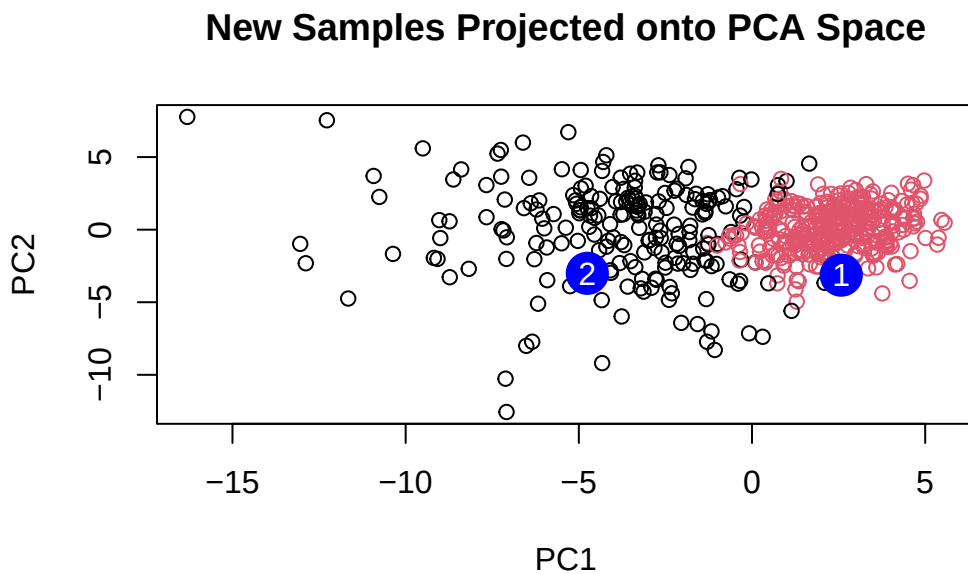
```

ylab = "PC2",
main = "New Samples Projected onto PCA Space")

points(npc[, 1], npc[, 2],
       col = "blue",
       pch = 16,
       cex = 3)

text(npc[, 1], npc[, 2],
     labels = c("1", "2"),
     col = "white")

```



Q16. Which new patient should be prioritized for follow-up?

Based on the PCA projection, patient/sample 2 should be prioritized for follow-up because it falls closer to the region associated with malignant samples.

Conclusion

This mini-project showed how PCA and hierarchical clustering can be used to explore biological data without using known labels during the analysis. PCA reduced the 30 original features into a smaller number of principal components while preserving most of the variation. Hierarchical clustering then grouped samples based on similarity. Overall, clustering after PCA produced a

clearer separation between malignant and benign samples than clustering on the original scaled data alone.

session info?

```
sessionInfo()
```

```
R version 4.5.3 (2026-03-11)
Platform: aarch64-apple-darwin20
Running under: macOS Sequoia 15.6.1
```

```
Matrix products: default
```

```
BLAS: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.0.dylib
```

```
LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;
```

```
locale:
```

```
[1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
```

```
time zone: America/Los_Angeles
```

```
tzcode source: internal
```

```
attached base packages:
```

```
[1] stats      graphics  grDevices  utils      datasets  methods    base
```

```
other attached packages:
```

```
[1] lubridate_1.9.5 forcats_1.0.1  stringr_1.6.0  dplyr_1.2.1
[5] purrr_1.2.2     readr_2.2.0    tidyr_1.3.2    tibble_3.3.1
[9] ggplot2_4.0.3   tidyverse_2.0.0
```

```
loaded via a namespace (and not attached):
```

```
[1] gtable_0.3.6      jsonlite_2.0.0    compiler_4.5.3    tidyselect_1.2.1
[5] scales_1.4.0      yaml_2.3.12       fastmap_1.2.0     R6_2.6.1
[9] labeling_0.4.3    generics_0.1.4    knitr_1.51        pillar_1.11.1
[13] RColorBrewer_1.1-3 tzdb_0.5.0        rlang_1.1.7       stringi_1.8.7
[17] xfun_0.57         S7_0.2.2          timechange_0.4.0  cli_3.6.5
[21] withr_3.0.2       magrittr_2.0.5    digest_0.6.39     grid_4.5.3
[25] rstudioapi_0.18.0 hms_1.1.4         lifecycle_1.0.5   vctrs_0.7.2
[29] evaluate_1.0.5    glue_1.8.0        farver_2.1.2      rmarkdown_2.31
[33] tools_4.5.3       pkgconfig_2.0.3   htmltools_0.5.9
```